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4. A group of students were using dice to model radioactive decay. There were 120 dice in total.

(a) (i) State the chance of any one of the dice landing with a 6 facing upward. [1]

....

(ii) Predict how many dice will land with a 6 facing up on the first throw. [1]

....

(iii) State what the students should do after throwing the 120 dice for the first time. [1]

....

(iv) Describe how they should continue the experiment. [2]

....

....

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(b) The table shows the results.

Throw	Number of sixes	Number of dice remaining
0	0	120
1	....	101
2	15	86
3	14	72
4	11	61
5	11	....

(i) **Complete** the table. [2]

(ii) How many dice remain after **one** half-life? .... [1]

(iii) Use the data in the table to estimate the half-life of the 'dice'. [1]

half-life = ... throws

(iv) Calculate the number of half-lives that will occur for the number of dice to fall from 120 to 15. [2]

number of half-lives = ...

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(c) Radioisotopes have many medical uses. One use is to sterilise blood products for transfusion.

The table below lists some radioisotopes.

Radioisotope	Half-life	Type of emissions
chromium-51	28 days	gamma
caesium-137	30 years	beta and gamma
bismuth-209	2×10^{19} years	alpha

The radioisotope used to sterilise blood products is caesium-137.

(i) Explain why caesium-137 is a better choice than chromium-51 for this purpose. [2]

....

....

(ii) Explain why caesium-137 is a better choice than bismuth-209 for this purpose. [2]

....

....

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